

LESSON 9.3

SUBTRACTING LINEAR EXPRESSIONS WITH RATIONAL COEFFICIENTS



LESSON INTRODUCTION

MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.

LEARNING TARGETS

- I can subtract linear expressions with rational coefficients.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.

WORKING TOWARD

MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.

ESSENTIAL QUESTIONS

- What is different about subtracting linear expressions from adding linear expressions?
- How do you subtract linear expressions?

LESSON NARRATIVE

Students extend their understanding of combining like terms from the previous lesson by exploring four different methods or strategies of subtracting linear expressions. Students analyze examples that start with integer coefficients and constants and build rational constants and coefficients in subtraction expressions with the fewest possible terms.

COMPONENT SUMMARY

9.3.1 WARM-UP (~5 MIN.)

Bell work adding linear expressions

9.3.3 COLLABORATION (10 MIN.)

Students collaborate to analyze student work using the distributive property and then relate this to algebra tiles

9.3.5 GUIDED PRACTICE (10 MIN.)

Students work with a partner to find the differences of linear expressions

9.3.7 WRAP-UP (~5 MIN.)

Combine like terms to simplify two linear expressions with rational coefficients

9.3.2 EXPLORATION (10 MIN.)

Students use algebra tiles to find the difference of linear expressions

9.3.4 GUIDED INSTRUCTION (10 MIN.)

Teacher models how to find the differences of linear expressions by adding the opposite or distributing a negative

9.3.6 YOUR TURN! (10 MIN.)

Students work independently to find the differences of linear expressions

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Additive inverse
- Distributive property

MATERIALS

- Algebra tiles
- (Optional) Notecards
- (Optional) Chart paper
- (Optional) Markers
- (Optional) Colored pencils

ADDITIONAL PREPARATION

- If using concrete algebra tiles, determine how many tiles are needed to model all of the examples in the lesson and organize student sets in a container or Ziplock bag.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may forget to distribute subtraction, or (-1) , to all terms in the parentheses (like in the 9.3.2 Exploration).
- Students may not simplify the expression completely or have trouble combining all of the like terms in the expressions.

THINGS TO CONSIDER

- When adding decimals, consider providing a model or anchor chart for rewriting problems vertically and line up the decimal places. When adding fractions, students may need to be reminded to obtain a common denominator.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

9.3.4 Guided Instruction provides an opportunity for students to Notice and Wonder about two methods for subtracting linear expressions. Students complete a table using the additive inverse method and applying the distributive property method. Engage the class in a Notice and Wonder routine to draw conclusions and connections between the methods.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

In 9.3.4 Guided Instruction, students apply two different methods to subtract linear equations. Use this as an opportunity for students to make connections between the two methods and understand how to solve linear subtraction in multiple ways.

DIFFERENTIATE INSTRUCTION

Consider providing Math Nation's On-Ramp video for operations with fractions as scaffolding for students when operating with mixed numbers or fractions greater than one (as seen in 9.3.4 Guided Practice and 9.3.6 Your Turn!). Avoid a previous learning target becoming a barrier to conceptual understanding for subtracting expressions by providing scaffolding.

9.3.1 WARM-UP: BELL WORK

DIFFERENTIATE INSTRUCTION

Consider having students circle or underline like terms before combining. This will provide a visual representation that can aid in understanding of what like terms are.



9.3.1 Warm-Up: Bell Work

Combine like terms.

$$1. (3y + 4.6) + (-5y - 3.25)$$

$$= (3y + (-5y)) + (4.6 - 3.25) = -2y + 1.35$$

$$2. \left(6\frac{3}{4}x - 7\right) + \left(-\frac{5}{2}x + 7\right)$$

$$= \left(\frac{27}{4}x + \left(-\frac{10}{4}x\right)\right) + ((-7) + 7)$$

$$= \frac{17}{4}x \text{ or } 4\frac{1}{4}x$$



9.3.2 Exploration: Subtracting Expressions with Algebra Tiles

1. Algebra tile representations are given below. Use the algebra tiles to find the difference.

On the lines, write the algebraic expression for each representation.	$(2x + 3) - (x - 5)$
Rearrange the tiles to show subtraction of like tiles together. Draw the representation.	$(x + 3) - (x - 5)$
If necessary, create any zero pairs needed for subtraction.	
Find the difference by subtracting (taking away) like tiles. Write the algebraic representation of the difference.	$x + 8$

9.3.2 EXPLORATION: SUBTRACTING EXPRESSIONS WITH ALGEBRA TILES

TEACHER MOVES

Make note of which students recognize that both x and -5 are being subtracted from $2x+3$ compared to students who only subtract x and then add -5 . Instead of formalizing and explaining the misconceptions, use this activity as an opportunity to instead notice the difference when comparing with peers.

Some students found the difference to be $x-2$ while others found it to be $x+8$. Why do you think that is? Which is correct?

What is being subtracted from $2x+3$?

9.3.3 COLLABORATION: SUBTRACTING EXPRESSIONS

TEACHER MOVES

This activity acts as a primer for 9.3.4 Guided Instruction. Monitor student responses while they collaborate and make note of any responses that you can use to build on when formalizing approaches in 9.3.4. Students should analyze and compare student responses before writing what students may have done incorrectly, as seeing the different examples will help them solidify their initial thoughts.

DIFFERENTIATE INSTRUCTION

Consider providing sentence stems for the written explanations, such as:

- “Noah’s answer is correct/incorrect. I know this because...”
- “The mistake he/she made was...”
- “It can be corrected by...”

This is a Collaboration, so prompt students to refine their responses with their partners.

ENRICHMENT

Consider probing questions for students or partners that are struggling to identify the differences in student work.

- Compare Noah and Jada’s work. What is the same? What is different?
- Compare Jada’s work to Andre’s work. What is the same? What is different?



9.3.3 Collaboration: Subtracting Expressions

- Noah, Lin, Jada, and Andre are rewriting the expression $(x + 8) - (4 - 9x)$ as an equivalent expression with fewer terms. Each student described their work.

Discuss with your partner which students made an error. Describe each student’s error and correct the error.

- Noah says, “I dropped the parentheses, combined like terms, and ended up with $-8x + 4$.”

$$\begin{aligned}(x + 8) - (4 - 9x) \\ x + 8 - 4 - 9x \\ -8x + 4\end{aligned}$$

Noah did not distribute (-1) to both terms. After distributing (-1) to both terms, the result is $x + 8 - 4 + 9x = (x + 9x) + (8 - 4) = 10x + 4$.

- Lin says, “I simplified inside the parentheses and ended up with $13x$.”

$$\begin{aligned}(x + 8) - (4 - 9x) \\ (8x) - (-5x) \\ 8x + 5x \\ 13x\end{aligned}$$

Lin combined terms that were not like terms. Lin should group the like terms together $(x + 9x) + (8 - 4)$ and then combine like terms to obtain $10x + 4$.

- Jada says, “I know that subtracting is adding the opposite, and ended up with $4 + 10x$.”

$$\begin{aligned}(x + 8) - (4 - 9x) \\ (x + 8) + (-4 + 9x) \\ x + 8 - 4 + 9x \\ 4 + 10x\end{aligned}$$

Jada has a correct answer. The explanation should include intermediate steps.

- Andre says, “I used the distributive property since there is an invisible (-1) in front of the second parentheses to distribute, but I ended up with $10x + 4$.”

$$\begin{aligned}(x + 8) - (4 - 9x) \\ (x + 8) - 1(4 - 9x) \\ x + 8 - 4 + 9x \\ 10x + 4\end{aligned}$$

Andre has a correct explanation and answer.

- When subtracting expressions, use the distributive property to add the opposite. With the algebra tile representations below, model finding the difference by adding the opposite.

On the lines, write the algebraic expression for each representation.	$(-3x + 4) - (2x - 7)$
Rewrite subtracting to adding the opposite by changing (or flipping) the sign of the 2 nd expression. Draw the representation.	$(-3x + 4) + (-2x + 7)$
Combine any zero pairs to find the sum. Write the sum.	$-5x + 11$



9.3.4 Guided Instruction: Subtracting Linear Expressions

1. To subtract linear expressions,
- add the *opposite* or additive inverse of each term in the expression.
- OR
- distribute a *negative one* to each term in the expression.
2. Complete the table below to show both methods.

Additive Inverse Method	Applying the Distributive Property
$(4x + 5) - (-2x + 6)$ $= (4x + 5) + (2x + (-6))$ $= (4x + 2x) + (5 + (-6))$ $= 6x + (-1)$ $= 6x - 1$	$(4x + 5) - (-2x + 6)$ $= 4x + 5 + 2x - 6$ $= (4x + 2x) + (5 + (-6))$ $= 6x + (-1)$ $= 6x - 1$

3. When adding and subtracting, parentheses are used to group expressions. Why is the use of parentheses especially important when subtracting?
- The use of parentheses ensures that the entire expression is subtracted. This means the negative 1 is distributed to all terms in the expression.*

9.3.4 GUIDED INSTRUCTION: SUBTRACTING LINEAR EXPRESSIONS

ENRICHMENT

- Consider using the following questions to extend student thinking and formalize their understanding of each approach following their written responses:
- How is the expression $(4x+5)-(-2x+6)$ different from the expression $4x+5--2x+6$?
 - What do the parentheses communicate about what is being subtracted?

9.3.5 GUIDED PRACTICE: SUBTRACTING LINEAR EXPRESSIONS WITH RATIONAL COEFFICIENTS

DIFFERENTIATE INSTRUCTION

Consider pulling a small group for students who would benefit from a model or think-aloud that highlights the usefulness of explicitly writing the coefficient -1 for better visual scaffolding of the distribution.

TEACHER MOVES

Notice the phrase “subtracted from” in question 2. Students might invert the order. Consider providing this prompt as a self-check for students:

I see some students found the difference to be $0.8x - 1.48$ while others found it to be $-0.8x - 1.48$.

- Which is correct?
- Which expression is being subtracted?



9.3.5 Guided Practice: Subtracting Linear Expressions with Rational Coefficients

1. Find the difference of each expression.

a. $(3y + 3.5) - (8.2y - 6)$ $= (3y + 3.5) - 8.2y + 6$ $= (3y + (-8.2y)) + (3.5 + 6)$ $= -5.2y + 9.5$	b. $(m - \frac{3}{8}) - (3\frac{1}{4} - m)$ $= (m - \frac{3}{8}) - 3\frac{1}{4} + m$ $= (m + m) + ((-\frac{3}{8}) + (-3\frac{1}{4}))$ $= 2m - 3\frac{5}{8}$
c. $(-2 + \frac{9}{2}y) - (5.3y - 5\frac{1}{2})$ $= (-2 + \frac{9}{2}y) - 5.3y + 5\frac{1}{2}$ $= (\frac{9}{2}y + (-5.3y)) + (-2 + 5\frac{1}{2})$ $= -0.8y + 3.5 \text{ or } -\frac{4}{5}y + 3\frac{1}{2}$	d. $(-7.5t - 3) - (7 + 3\frac{1}{8}t)$ $= (-7.5t - 3) - 7 - 3\frac{1}{8}t$ $= -7.5t - 3\frac{1}{8}t + (-3 + -7)$ $= -7.5t - 3\frac{1}{8}t + (-10)$ $= -10\frac{5}{8}t - 10$

2. What is the result when $-3x + 4.48$ is subtracted from $-2.2x + 3$?

- Write the algebraic expression.
 $(-2.2x + 3) - (-3x + 4.48)$
- Rewrite the expression without parentheses.
 $-2.2x + 3x + 3 + (-4.48)$
- Simplify the expression by rewriting it using the fewest terms possible.
 $0.8x - 1.48$

**9.3.6 Your Turn!**

1. Find the difference of each expression in simplest form.

a. $(-2h + 8) - (h + 10)$ $= (-2h + 8) - h - 10$ $= (-2h + (-h)) + (8 + (-10))$ $= -3h - 2$	b. $(\frac{7}{2}x + \frac{4}{3}) - (3 - \frac{9}{8}x)$ $= (\frac{7}{2}x + \frac{4}{3}) - 3 + \frac{9}{8}x$ $= (\frac{7}{2}x + \frac{9}{8}x) + (\frac{4}{3} + (-3))$ $= \frac{37}{8}x - 1\frac{2}{3} \text{ or } 4\frac{5}{8}x - 1\frac{2}{3}$
c. $(-4.4y - 9) - (5 - 2.1y)$ $= (-4.4y - 9) - 5 + 2.1y$ $= (-4.4y + 2.1y) + (-9 + (-5))$ $= -2.3y - 14$	d. $(2\frac{2}{3}n + 1) - (-\frac{1}{4}n - 6)$ $= (2\frac{2}{3}n + 1) + \frac{1}{4}n + 6$ $= (2\frac{2}{3}n + \frac{1}{4}n) + (1 + 6)$ $= 2\frac{11}{12}n + 7$

2. What is the result when $\frac{5}{8}x + \frac{4}{3}$ is subtracted from

$$\frac{5}{4}x - \frac{31}{6}$$

$$= (\frac{5}{4}x - \frac{31}{6}) - \frac{5}{8}x - \frac{4}{3}$$

$$= (\frac{5}{4}x + (-\frac{5}{8}x)) + (-\frac{31}{6} + (-\frac{4}{3}))$$

$$= \frac{5}{8}x - \frac{39}{6} = \frac{5}{8}x - 6\frac{1}{2}$$

**9.3.7 Wrap-Up: Ticket Out the Door**

Write the following expressions using the fewest possible terms.

1. $(5x - 13\frac{1}{4}) - (4\frac{1}{2} - 5x)$
- $$= 5x + 5x - 13\frac{1}{4} - 4\frac{1}{2}$$
- $$= 10x - 17\frac{3}{4}$$
2. $(0.7 - 2.4x) + (-5.32 + 4\frac{1}{10}x)$
- $$= 0.7 - 5.32 - 2.4x + 4\frac{1}{10}x$$
- $$= -4.62 + 1.7x$$

9.3.6 YOUR TURN!**DIFFERENTIATE INSTRUCTION**

This benchmark does include any rational number as a coefficient. For students who need additional entry points to engage with the content when fractions and decimals are involved, consider providing Math Nation's On-Ramp video on adding and subtracting fractions.

TEACHER MOVES

Note perhaps the most common misconception in subtracting can manifest here. Students may switch the order of the expressions based on whether they recognize and understand what “subtract from” means about the order of the expressions. If many students make this error, consider reviewing as a whole group to clarify this important misconception.

9.3.7 WRAP-UP: TICKET OUT THE DOOR**TEACHER MOVES**

Students may get $0x - 17\frac{3}{4}$ or $-17\frac{3}{4}$ for the first linear expression. Remind students that writing the coefficient of -1 in front of the second parenthesis can help them remember to distribute.

DIFFERENTIATE INSTRUCTION

While the lesson's learning target is focused on subtracting linear expressions, this ticket out the door is intended to be used as formative assessment for both adding and subtracting linear expressions following the first three lessons of the unit. Use the informal data gathered from this formative assessment to plan for adjustments or to identify small groups who may need support in lesson 4.